

Grade 8 Science
Optics Test Checklist

1. Electromagnetic spectrum: Know what the EM spectrum is, what an electromagnetic wave is, what ROY G BIV means and be aware of the following properties of the EM spectrum (different types of EM energy have different wavelengths, the relationship between frequency and wavelength, and which type of waves, shorter or longer wavelength, have the most and least energy)
2. Know the difference between transparent, translucent and opaque
3. Know what the normal, optical centre, and the principal axis line are
4. **Reflection:**
 - A. Know what reflection is and the difference between real and virtual images in mirrors
 - B. Know what mirrors are, be aware of the different types of mirrors (plane, convex, concave)
 - C. Know how light rays behave when they hit plane, convex, and concave mirrors
 - D. Know how images behave in plane, convex, and concave mirrors
 - E. Understand how the behavior of the image formed by the mirror and the behavior of the light rays are related to one another
 - F. Understand the law of reflection for plane mirrors: angle of incidence = angle of reflection
 - G. Know the difference between specular and diffuse reflection
 - H. Know what converging and diverging light rays mean and know what a focal point is
 - I. Know what scattering of light is, and how it contributes to beam of light being visible in chalk dust and to the sky being blue
5. **Refraction:**
 - A. Know what refraction is and what causes it
 - B. Know how light rays behave when they pass through convex and concave lenses
 - C. Know how to draw ray diagrams for convex and concave lenses in order to determine if the image is real or virtual, smaller, larger, or the same size as the original object, and upright or inverted
 - D. Know what dispersion of light is (white light splitting into the colors of ROY G BIV through a triangular prism) and what causes it

6. **The Eye:**

- A. Know the location and function of the cornea, pupil, iris, sclera, optic nerve, lens, retina, and know that the eye contains fluid and a gel like substance to help it keep its shape
- B. Know what rods and cones are, where they are located, and what they do
- C. Know how eye the forms an image in people with normal eyesight
- D. Know what nearsighted and farsighted mean, what causes them, and how to correct them